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Semiconductor device and power conversion device

Abstract

A highly reliable semiconductor device and power conversion device are obtained. In the semiconductor device, a semiconductor element is mounted on a main surface of a base member. A case includes a side wall surrounding the base member. A support portion is formed on an inner peripheral surface of the side wall on a side of the base member. At least a part of the support portion is in contact with a contact region located inside an outer peripheral end of wiring board on a surface of the wiring board located on the side of the semiconductor element. A sealing member seals the semiconductor element and the wiring board. A terminal member is connected to a region of the wiring board closer to the outer peripheral end than the contact region. The terminal member includes an end portion protruding from a surface of the sealing member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is based on PCT filing PCT/JP2020/027383, filed Jul. 14, 2020, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a semiconductor device and a power conversion device.

BACKGROUND ART

(3) Conventionally, as the semiconductor device used for the power conversion device, there has been known a semiconductor device in which a wiring board is stacked on a semiconductor element, and the semiconductor element and the wiring board are sealed with a sealing member (for example, see Japanese Patent Laying-Open No. 2009-81328).

CITATION LIST

Patent Literature

(4) PTL 1: Japanese Patent Laying-Open No. 2009-81328

SUMMARY OF INVENTION

Technical Problem

(5) In the conventional semiconductor device, a terminal member such as a lead terminal connecting the wiring board and the outside is disposed so as to extend from the wiring board to the outside of the sealing member. In this case, there is a possibility that moisture enters the semiconductor device from an external environment along an interface between the terminal member and the sealing member. When the moisture intrudes into the semiconductor device as described above, the moisture may cause degradation or damage of the semiconductor element, leading to degradation of reliability of the semiconductor device.

(6) The present disclosure has been made to solve the above problems, and an object of the present disclosure is to provide a semiconductor device and a power conversion device with high reliability.

Solution to Problem

(7) A semiconductor device according to the present disclosure includes a base member, a semiconductor element, a case, a wiring board, a sealing member, and a terminal member. The base member includes a main surface. The semiconductor element is mounted on the main surface of the base member. The case includes a side wall surrounding the base member. A support portion is formed on an inner peripheral surface of the side wall on a base member side. The wiring board is disposed at a position overlapping with the semiconductor element when viewed from a direction perpendicular to the main surface of the base member. An electronic component is mounted on the wiring board. At least a part of the support portion is in contact with a contact region located inside an outer peripheral edge of the wiring board on a surface of the wiring board located on a semiconductor element side. The sealing member is disposed inside the case. The sealing member seals the semiconductor element and the wiring board. The terminal member is connected to a region on an outer peripheral edge side of the contact region in the wiring board. The terminal member includes an end portion protruding from a surface of the sealing member.

(8) A power conversion device according to the present disclosure includes a main conversion circuit and a control circuit. The main conversion circuit includes the semiconductor device, and converts and outputs input power. The control circuit outputs a control signal controlling the main conversion circuit to the main conversion circuit.

Advantageous Effects of Invention

(9) According to the above, the highly reliable semiconductor device and power conversion device can be obtained.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic plan view illustrating a semiconductor device according to a first embodiment.
- (2) FIG. 2 is a schematic sectional view taken along a line II-II in FIG. 1.
- (3) FIG. 3 is a schematic sectional view taken along a line III-III in FIG. 1.
- (4) FIG. 4 is a schematic plan view illustrating a modification of the semiconductor device of the first embodiment.
- (5) FIG. 5 is a schematic plan view illustrating a semiconductor device according to a second embodiment.
- (6) FIG. 6 is a schematic plan view illustrating a positional relationship between a sealing member inlet port and an insulating member in the semiconductor device in FIG. 5.
- (7) FIG. 7 is a schematic sectional view taken along a line VII-VII in FIG. 5.
- (8) FIG. 8 is a sectional plan view illustrating a semiconductor device according to a third embodiment.
- (9) FIG. 9 is a schematic plan view illustrating a positional relationship between a sealing member inlet port and a semiconductor element in the semiconductor device in FIG. 8.
- (10) FIG. 10 is a sectional plan view illustrating a semiconductor device according to a fourth embodiment.
- (11) FIG. 11 is a schematic sectional view taken along a line XI-XI in FIG. 10.
- (12) FIG. 12 is a schematic sectional view illustrating a first modification of the semiconductor device of the fourth embodiment.
- (13) FIG. 13 is a schematic plan view illustrating a second modification of the semiconductor device of the fourth embodiment.
- (14) FIG. 14 is a sectional plan view illustrating a semiconductor device according to a fifth embodiment.
- (15) FIG. 15 is a schematic sectional view taken along a line XV-XV in FIG. 14.
- (16) FIG. 16 is a schematic sectional view taken along a line XVI-XVI in FIG. 14.
- (17) FIG. 17 is a block diagram illustrating a configuration of a power conversion system to which a power conversion device according to a sixth embodiment is applied.

DESCRIPTION OF EMBODIMENTS

(18) Hereinafter, embodiments of the present disclosure will be described. The same components are denoted by the same reference numerals, and a repetitive description will be omitted. The following drawings are schematic and may not reflect an exact size of an illustrated component.

First Embodiment

(19) <Configuration of Semiconductor Device>

(20) FIG. 1 is a schematic plan view illustrating a semiconductor device according to a first embodiment. FIG. 2 is a schematic sectional view taken along a line II-II in FIG. 1. FIG. 3 is a schematic sectional view taken along a line III-III in FIG. 1. The semiconductor device in FIGS. 1 to 3 mainly includes a base member 3, a semiconductor element 4, a case 6, a wiring board 11, a

sealing member **14**, and a terminal member **9**.

(21) Base member **3** includes a heat dissipation layer **1**, an insulating layer **25**, and a conductor layer **24**. Insulating layer **25** is laminated on heat dissipation layer **1** made of, for example, aluminum, copper, or the like. Conductor layer **24** as an upper electrode is disposed on insulating layer **25**. An upper surface of conductor layer **24** configures a main surface **3a** of base member **3**. Insulating layer **25** is formed so as to cover the upper surface of heat dissipation layer **1**. Insulating layer **25** can insulate only heat dissipation layer **1** and conductor layer **24** from each other, and may be formed only on a part of the upper surface of heat dissipation layer **1**. Conductor layer **24** is formed so as to cover a part of the upper surface of insulating layer **25**. A planar shape of base member **3** is a quadrangular shape. The planar shape of base member **3** may be a polygonal shape other than the quadrangular shape, or be a circular shape.

(22) A material of insulating layer **25** is not particularly limited, but may be a resin material in which at least one of a fine particle and a filler is dispersed. For example, at least one of the fine particle and the filler may be formed of an inorganic ceramic material such as silica (SiO_2), alumina (Al_2O_3), aluminum nitride (AlN), boron nitride (BN), silicon nitride (Si_3N_4), diamond (C), silicon carbide (SiC), or boron oxide (B_2O_3). At least one of the fine particle and the filler may be formed of a resin material such as a silicone resin or an acrylic resin. The resin material in which at least one of the fine particle and the filler is dispersed has an electrical insulation property. The resin material in which at least one of the fine particle and the filler is dispersed is not particularly limited, but may be an epoxy resin, a polyimide resin, a silicone resin, or an acrylic resin.

(23) Semiconductor element **4** is mounted on main surface **3a** that is the upper surface of conductor layer **24** of base member **3**. A plurality of semiconductor elements **4** are mounted on main surface **3a** of base member **3**. Semiconductor element **4** may be a power semiconductor element such as an insulated gate bipolar transistor (IGBT) or a metal oxide semiconductor field effect transistor (MOSFET), or be a diode such as a freewheel diode. A material constituting semiconductor element **4** may be silicon (Si), or a wide band gap semiconductor material such as silicon carbide (SiC), gallium nitride (GaN), or diamond. A required number of semiconductor elements **4** are fixed to the conductor layer **24** according to the use of the semiconductor device. In the plurality of semiconductor elements **4**, the same type of semiconductor elements **4** may be included, or different types of semiconductor elements **4** may be included. The materials constituting the plurality of semiconductor elements **4** may be different from each other or may be the same material.

(24) Semiconductor element **4** is bonded to conductor layer **24** by a bonding material **2** made of solder or the like. Any conductive material can be used as bonding material **2** in addition to the solder described above, but for example, silver or a silver alloy may be used.

(25) Case **6** is disposed so as to surround an outer periphery of base member **3**. Case **6** is connected to an outer peripheral portion of base member **3**. Case **6** surrounds base member **3** and includes a side wall **61** extending in a direction intersecting with main surface **3a** of base member **3**. An opening is formed on an upper side of side wall **61** located on a side opposite to a lower side connected to base member **3**. A size of the opening may be larger than a size of base member **3**. A planar shape of the opening may be the same as a planar shape of base member **3**. On the lower side of side wall **61**, a part of case **6** is connected to base member **3** so as to cover the outer peripheral portion of base member **3**. Specifically, a part of case **6** is connected to the outer peripheral portion of insulating layer **25** of base member **3**. A part of case **6** is connected to insulating layer **25** of base member **3** and the outer peripheral end face of heat dissipation layer **1**. Case **6** and base member **3** may be fixed by an adhesive or the like, and fixed by other means such as a screw.

(26) A support portion **15** is formed on an inner peripheral surface **61a** of side wall **61** on the side of base member **3**. In a direction perpendicular to main surface **3a** of base member **3**, support

portion **15** is disposed spaced apart from semiconductor element **4**. In the semiconductor device in FIGS. **1** to **3**, support portions **15** are formed on a pair of opposing side walls among side walls **61**. Support portion **15** is preferably formed so as to surround semiconductor element **4** from at least two directions. Preferably, support portion **15** is formed so as to surround semiconductor element **4** from three directions. More preferably, support portion **15** is formed so as to surround semiconductor element **4** from all directions. A resin having the electrical insulation property such as an epoxy resin, a polyimide resin, an acrylic resin, or a polyphenylene sulfide (PPS) resin may be used as a material constituting case **6**.

(27) Wiring board **11** is disposed at a position overlapping with semiconductor element **4** when viewed from the direction perpendicular to main surface **3a** of base member **3**. That is, wiring board **11** is disposed on base member **3** and spaced apart from semiconductor element **4**. An electronic component **10** is mounted on wiring board **11**. Electronic component **10** is disposed on each of a surface **11a** of wiring board **11** located on the side of semiconductor element **4** and the upper surface located on the opposite side of surface **11a**. In addition, a through-hole **12** reaching surface **11a** from the upper surface is made in wiring board **11**. A plurality of through-holes **12** are made in wiring board **11**. For example, a diameter of through-hole **12** is less than or equal to 1 mm. In this case, even when the inside of through-hole **12** is not filled with sealing member **14**, there is no significant influence on the effect of preventing wiring board **11** from reaching semiconductor element **4** by moisture or the like. Wiring board **11** is supported by support portion **15** such that surface **11a** of wiring board **11** is parallel to main surface **3a** of base member **3**. The size of surface **11c** of wiring board **11** on which electronic component **10** is mounted may be larger than the size of conductor layer **24**.

(28) Terminal member **9** is connected to wiring board **11**. A root portion of terminal member **9** is connected to the outer peripheral portion of wiring board **11**. Terminal member **9** is preferably disposed at a position close to side wall **61** of case **6**. An end portion **9a** of terminal member **9** is disposed so as to protrude from the upper surface of sealing member **14**. An internal connection terminal **8** is connected to wiring board **11**. As illustrated in FIGS. **1** and **3**, internal connection terminal **8** is connected to wiring board **11** inside terminal member **9** away from side wall **61** of case **6**. Internal connection terminal **8** and wiring board **11** are electrically connected by a bonding material such as solder. In addition, terminal member **9** and wiring board **11** are also electrically connected by the bonding material such as solder. The bonding material may be protected by covering a connection portion between internal connection terminal **8** and wiring board **11** and a connection portion between terminal member **9** and wiring board **11** with resin. A lower end portion of internal connection terminal **8** opposite to an upper end portion connected to wiring board **11** is fixed to case **6**. The lower end portion of internal connection terminal **8** is fixed to a portion covering the outer peripheral portion of base member **3** in case **6**.

(29) The outer peripheral portion of surface **11a** of wiring board **11** is in contact with support portion **15**, whereby wiring board **11** is supported by case **6**. A support surface **15c** that is the upper surface of support portion **15** is in contact with the outer peripheral portion of surface **11a** of wiring board **11**. Support portion **15** disposed under a region of wiring board **11** to which terminal member **9** is connected includes an extending portion extending from inner peripheral surface **61a** of side wall **61** away from terminal member **9** and the outer peripheral portion of wiring board **11** and a contact portion extending from a distal end of the extending portion toward the side of wiring board **11**. An upper end face of the contact portion is support surface **15c**. Support surface **15c** is in contact with surface **11a** of wiring board **11**. The region of wiring board **11** in contact with support surface **15c** of support portion **15** is a contact region **11aa**. That is, at least apart of support portion **15** is in contact with contact region **11aa** located inside an outer peripheral edge **11b** of wiring board **11** on surface **11a** located on the side of semiconductor element **4** of wiring board **11**. In other words, terminal member **9** is connected to the region of wiring board **11** closer to outer peripheral edge **11b** than contact region **11aa**. As a result, terminal member **9** is in a state of being

received by support portion **15**. A sectional shape of support portion **15** located directly below terminal member **9** may be an L-shape as illustrated in FIG. **3**.

(30) A recessed portion **15b** facing the region of wiring board **11** to which terminal member **9** is connected is formed in support portion **15** located under terminal member **9**. As a result, the outer peripheral portion of wiring board **11** to which terminal member **9** is connected is partitioned from the region on the side of semiconductor element **4** by support portion **15**.

(31) Support portion **15** is preferably in contact with as many portions as possible of the outer peripheral portion of wiring board **11**. For example, support portion **15** may be formed on the entire circumference of inner peripheral surface **61a** of side wall **61** of case **6**. As illustrated in FIG. **1**, support portion **15** is preferably formed on at least two opposing side walls **61**.

(32) Although the lower portion of terminal member **9** penetrates wiring board **11** in FIG. **3**, when terminal member **9** is connected to wiring board **11** without penetrating wiring board **11**, support portion **15** may be disposed immediately below terminal member **9**. For example, support portion **15** may be in contact with surface **11a** of wiring board **11** in the region located immediately below terminal member **9**. When a protrusion amount of terminal member **9** from surface **11a** of wiring board **11** is relatively small while terminal member **9** penetrates wiring board **11**, for example, a recess may be formed in support surface **15c** of support portion **15** and the end portion of terminal member **9** may be received in the recess. Case **6** in the semiconductor device in FIG. **3** includes support portion **15** having a left-right asymmetric shape according to the disposition of wiring board **11** and terminal member **9**, but the shape of support portion **15** may be a left-right symmetric shape in FIG. **3**.

(33) One of semiconductor elements **4** is electrically connected to internal connection terminal **8** by a conductive wire **5**. An external connection terminal **7** is embedded in case **6**. The lower portion of external connection terminal **7** is exposed to the surface of case **6** in the region overlapping with base member **3**. The upper portion of external connection terminal **7** protrudes from a top surface of side wall **61** of case **6**. Conductor layer **24** to which semiconductor element **4** is fixed and the lower portion of external connection terminal **7** are connected by conductive wire **5**. Semiconductor element **4** is electrically connected to external connection terminal **7** by conductive wire **5**. The material of conductive wire **5** is not particularly limited, but a metal material such as copper or aluminum can be used.

(34) Sealing member **14** is disposed inside case **6**. Sealing member **14** seals conductor layer **24** of base member **3**, semiconductor element **4**, wiring board **11**, electronic component **10**, internal connection terminal **8**, and conductive wire **5**. Sealing member **14** has the electrical insulation property and insulates the sealed members. A thickness **T1** of sealing member **14** is large enough to allow both semiconductor element **4** and electronic component **10** mounted on wiring board **11** to be embedded.

(35) In the semiconductor device, water vapor transmission rates of the materials constituting wiring board **11** and case **6** may be lower than a water vapor transmission rate of sealing member **14**. In this case, because the region where semiconductor element **4** is disposed is surrounded by wiring board **11** and case **6**, the possibility that the moisture or the like reaches semiconductor element **4** can be decreased. Because the contact portion between support portion **15** and wiring board **11** serves as a seal portion, support portion **15** is preferably disposed so as to surround the region from at least two directions. The water vapor transmission rate is measured by, for example, a moisture sensor method defined in JIS standard K7126-1.

(36) The material of sealing member **14** is preferably an epoxy resin, but is not limited thereto, and for example, an insulating resin such as a silicone resin, a urethane resin, a polyimide resin, a polyamide resin, or an acrylic resin may be used. The material constituting sealing member **14** may be an insulating resin material in which the fine particles or fillers improving strength and thermal conductivity of sealing member **14** are dispersed. For example, the fine particle or filler improving the strength and thermal conductivity of sealing member **14** may be formed of an inorganic ceramic

material such as silica (SiO.sub.2), alumina (Al.sub.2O.sub.3), aluminum nitride (AlN), boron nitride (BN), silicon nitride (Si.sub.3N.sub.4), diamond (C), silicon carbide (SiC), or boron oxide (B.sub.2O.sub.3).

(37) <Method for Manufacturing Semiconductor Device>

(38) A method for manufacturing the semiconductor device in FIGS. 1 to 3 will be described.

(39) First, a process (S10) of preparing members constituting a semiconductor device is performed. In this process (S10), members constituting the semiconductor device are prepared, members including base member 3, semiconductor element 4, case 6, wiring board 11 on which electronic component 10 is mounted, and sealing member 14.

(40) Subsequently, a process (S20) of bonding semiconductor element 4 to base member 3 is performed. In this process (S20), solder as bonding material 2 and semiconductor element 4 are disposed on base member 3. Thereafter, the solder is melted by reflow heating and then solidified to bond semiconductor element 4 to base member 3.

(41) Subsequently, a process (S30) of bonding case 6 to base member 3 is performed. In this process (S30), an adhesive (not illustrated) is uniformly applied to the outer peripheral portion of base member 3 to which semiconductor element 4 is soldered. Case 6 is brought into contact with the portion to which the adhesive is applied, and case 6 and base member 3 adhere to each other. Here, as illustrated in FIGS. 1 and 3, case 6 includes an external connection terminal 7 and internal connection terminal 8. External connection terminal 7 is drawn out of the semiconductor device through the inside of case 6. As illustrated in FIG. 3, internal connection terminal 8 is disposed so as to extend upward (along side wall 61) from the lower portion of case 6.

(42) Subsequently, a process (S40) of forming a circuit is performed. In this process (S40), semiconductor element 4, conductor layer 24 of base member 3, external connection terminal 7, internal connection terminal 8, and the like are electrically connected using conductive wire 5 to form the circuit.

(43) Subsequently, a process (S50) of disposing wiring board 11 is performed. In this process (S50), wiring board 11 is disposed inside case 6 such that support surface 15c of support portion 15 formed on side wall 61 of case 6 comes into contact with the outer peripheral portion of lower surface 11a of wiring board 11. Wiring board 11 is disposed at a position where wiring board 11 and the distal end of internal connection terminal 8 can be bonded. At this time, the contact area between support surface 15c and wiring board 11 is preferably as wide as possible. Support surface 15c is preferably in contact with the entire periphery of the outer peripheral portion of wiring board 11. As illustrated in at least FIGS. 1 and 3, preferably support surface 15c and wiring board 11 is in contact with each other on two opposing sides of the outer peripheral portion of wiring board 11.

(44) Subsequently, a soldering process (S60) is performed. In this process (S60), the lower portion of terminal member 9 for external connection is disposed at a position bonded to wiring board 11. Then, wiring board 11 and internal connection terminal 8 are electrically connected by soldering. In addition, wiring board 11 and terminal member 9 are electrically connected by soldering.

(45) Subsequently, a sealing process (S70) is performed. In this process (S70), the sealing resin as sealing member 14 is injected into the region below wiring board 11 from the portion where wiring board 11 is not supported by support portion 15. Furthermore, sealing member 14 is injected into case 6 so as to embed all electronic components 10 of wiring board 11. As a result, sealing member 14 is disposed so as to embed bonding material 2 such as solder, conductor layer 24, semiconductor element 4, conductive wire 5, internal connection terminal 8, electronic component 10, wiring board 11, support portion 15 formed on inner peripheral surface 61a of side wall 61 of case 6, and the connection portion between support portion 15 and wiring board 11. As a result, the semiconductor device in FIGS. 1 to 3 can be obtained.

Advantageous Effect

(46) The semiconductor device of the present disclosure includes base member 3, semiconductor element 4, case 6, wiring board 11, sealing member 14, and terminal member 9. Base member 3

include main surface **3a**. Semiconductor element **4** is mounted on main surface **3a** of base member **3**. Case **6** includes side wall **61** surrounding base member **3**. A support portion **15** is formed on an inner peripheral surface **61a** of side wall **61** on the side of base member **3**. Wiring board **11** is disposed at a position overlapping with semiconductor element **4** when viewed from the direction perpendicular to main surface **3a** of base member **3**. An electronic component **10** is mounted on wiring board **11**. At least a part of support portion **15** is in contact with contact region **11aa** located inside outer peripheral edge **11b** of wiring board **11** on surface **11a** located on the side of semiconductor element **4** of wiring board **11**. Sealing member **14** is disposed inside case **6**. Sealing member **14** seals semiconductor element **4** and wiring board **11**. Terminal member **9** is connected to the region of wiring board **11** closer to outer peripheral edge **11b** than contact region **11aa**. Terminal member **9** includes end portion **9a** protruding from the surface of sealing member **14**.

(47) In this way, wiring board **11** sealed by sealing member **14** is disposed on semiconductor element **4**, a path length through which a substance such as the moisture or sulfur gas enters from the outside from the surface of sealing member **14** to semiconductor element **4** can be increased as compared with the case where wiring board **11** does not exist. In addition, terminal member **9** is connected to the outer peripheral side of contact region **11aa** with which support portion **15** is in contact in wiring board **11**, so that the possibility that the substance such as the moisture reaches semiconductor element **4** from the outside through the contact interface between terminal member **9** and sealing member **14** can be decreased as compared with the case where the terminal member **9** extends from the vicinity of the semiconductor element **4** to the outside of the sealing member **14**.

(48) Here, a case where a substance such as moisture reaches the outer peripheral portion of the wiring board **11** from the outside through the contact interface between the terminal member **9** and the sealing member **14** is considered. In this case, because support portion **15** is in contact with contact region **11aa** of wiring board **11** between the outer peripheral portion and semiconductor element **4**, the contact interface between wiring board **11** and support portion **15** and sealing member **14** is formed. The contact interface traps the substance such as the moisture reaching the outer peripheral portion of wiring board **11**, and as a result, the possibility that the substance such as the moisture reach semiconductor element **4** can be decreased. As a result, the reliability of the semiconductor device can be improved.

(49) In the semiconductor device, the water vapor transmission rates of the materials constituting wiring board **11** and case **6** may be smaller than the water vapor transmission rate of sealing member **14**.

(50) In this case, because wiring board **11** having the relatively low water vapor transmission rate is embedded on semiconductor element **4** inside sealing member **14**, wiring board **11** acts as the member that prevents water from entering from the outside toward semiconductor element **4** from the upper surface of sealing member **14** or a member that increases a water entry path length. Thus, the possibility that the substance such as the water reaches semiconductor element **4** can be decreased.

(51) In the semiconductor device, base member **3** may include conductor layer **24** to which semiconductor element **4** is connected. The size of surface **11c** of wiring board **11** on which electronic component **10** is mounted may be larger than the size of conductor layer **24**.

(52) In this case, wiring board **11** disposed in sealing member **14** can sufficiently increase the length of the entry path of the substance such as the water from the upper surface of sealing member **14**.

(53) In the semiconductor device, at least a part of support portion **15** may include recessed portion **15b** facing the region of wiring board **11** to which terminal member **9** is connected. In this case, recessed portion **15b** of support portion **15** can form the seal portion that surrounds the region of wiring board **11** to which terminal member **9** is connected. Therefore, the substance such as the water can be prevented from entering the side of semiconductor element **4** through a bonded interface between terminal member **9** and sealing member **14** by the seal portion.

(54) In the semiconductor device, base member **3** may include heat dissipation layer **1**. The material constituting heat dissipation layer **1** may be metal. For example, in the semiconductor device, base member **3** may include conductor layer **24** and heat dissipation layer **1**. Semiconductor element **4** is connected to conductor layer **24**. Heat dissipation layer **1** is connected to conductor layer **24** with insulating layer **25** interposed therebetween. The material constituting heat dissipation layer **1** may be metal.

(55) In this case, because base member **3** includes heat dissipation layer **1**, the configuration of the semiconductor device can be simplified as compared with the case where the heat dissipation layer is connected to base member **3** as a separate member.

(56) In the semiconductor device, support portion **15** may be in contact with wiring board **11** in at least two directions of the outer peripheral portion of wiring board **11**. In this case, the contact portion between support portion **15** and wiring board **11** acts as the seal portion, and the possibility that the substance such as the water enters semiconductor element **4** from the upper surface of sealing member **14** can be decreased.

(57) In addition, as illustrated in FIGS. **1** and **3**, preferably the component such as electronic component **10** is not exposed on the upper surface of sealing member **14** inside a support surface end portion **15ca** located on the distal end side of support portion **15**. In this case, the connection interface between electronic component **10** and sealing member **14** is not exposed on the upper surface of sealing member **14** inside support surface end portion **15ca**. Accordingly, generation of the problem that atmospheric gas such as water vapor and sulfur gas enters the inside of the semiconductor device through the connection interface can be prevented. In addition, wiring board **11** is disposed inside sealing member **14** and above semiconductor element **4**, so that the entry path length of the gas that can be formed from the upper surface of sealing member **14** to semiconductor element **4** can be lengthened. As a result, the possibility that the moisture or the like reaches semiconductor element **4** to generate a defect can be decreased without requiring a special additional member, and a highly reliable semiconductor device can be obtained.

(58) <Configuration and Advantageous Effect of Modification>

(59) FIG. **4** is a schematic plan view illustrating a modification of the semiconductor device of the first embodiment. FIG. **4** corresponds to FIG. **1**. The semiconductor device in FIG. **4** basically includes the same configuration as the semiconductor device in FIGS. **1** to **3**, but the configuration of support portion **15** is different from that of the semiconductor device in FIGS. **1** to **3**.

Specifically, a notch **15a** in which a part of support portion **15** is missing is formed in the semiconductor device in FIG. **4**. A distance **L1** from notch **15a** to semiconductor element **4** is preferably larger than thickness **T1** of sealing member **14**.

(60) Even with such the configuration, the same effects as those of the semiconductor device in FIGS. **1** to **3** can be obtained. In addition, when sealing member **14** is supplied into case **6** in the method for manufacturing the semiconductor device, sealing member **14** can be supplied to the region between wiring board **11** and semiconductor element **4** through notch **15a**. In addition, distance **L1** from notch **15a** to semiconductor element **4** in planar view (when viewed from the direction perpendicular to main surface **3a** of base member **3**) is larger than thickness **T1** (see FIG. **1**) of sealing member **14**, so that an increase in the possibility that the substance such as the water enters semiconductor element **4** from the upper surface of sealing member **14** due to the formation of notch **15a** can be prevented.

Second Embodiment

(61) <Configuration of Semiconductor Device>

(62) FIG. **5** is a schematic plan view illustrating a semiconductor device according to a second embodiment. FIG. **6** is a schematic plan view illustrating a positional relationship between a sealing member inlet port and an insulating member in the semiconductor device in FIG. **5**. FIG. **6** schematically illustrates disposition of base member **3**, semiconductor element **4**, and conductive wire **5** located below wiring board **11** in FIG. **5**. FIG. **7** is a schematic sectional view taken along a

line VII-VII in FIG. 5. FIG. 5 corresponds to FIG. 1, and FIG. 7 corresponds to FIG. 3.

(63) The semiconductor device in FIGS. 5 to 7 basically has the same configuration as the semiconductor device in FIGS. 1 to 3, but is different from the semiconductor device in FIGS. 1 to 3 in that sealing member inlet port 13 is formed in wiring board 11 and in the configuration of support portion 15. Here, a hole that penetrates wiring board 11 from surface 11a, which is the back surface, to the upper surface does not insert the electronic component, but has a diameter of at least 1 mm is defined as sealing member inlet port 13. In planar view in FIGS. 5 and 6 (when viewed from the direction perpendicular to main surface 3a of base member 3), sealing member inlet port 13 is formed at the position overlapping with case 6. More specifically, sealing member inlet port 13 in wiring board 11 is formed at the position overlapping with case surface 6a that is the upper surface of the portion extending on the outer peripheral portion of base member 3 in case 6. That is, sealing member inlet port 13 is disposed at the position not overlapping with conductor layer 24 of base member 3 in planar view.

(64) Although it is ideal that sealing member inlet port 13 exists at the position farthest from conductor layer 24, at least a distance L2 between sealing member inlet port 13 and conductor layer 24 is preferably greater than or equal to thickness T1 (see FIG. 1) of sealing member 14 in planar view of FIG. 6. As described above, sealing member inlet port 13 is not provided on conductor layer 24, but sealing member inlet port 13 is provided at the position separated from conductor layer 24 by at least thickness T1 of sealing member 14, so that gas such as water vapor or sulfur gas can be prevented from reaching semiconductor element 4 from the upper surface of sealing member 14.

(65) In the semiconductor device described above, sealing member inlet port 13 is formed in wiring board 11, so that sealing member 14 can be injected from sealing member inlet port 13 to the lower side of wiring board 11. Accordingly, support portion 15 is formed so as to surround the entire periphery of wiring board 11 so as to be in contact with the entire outer periphery of wiring board 11. That is, support portion 15 is formed in an annular shape in planar view so as to go around the inner peripheral surface of the side wall of case 6.

(66) <Method for Manufacturing Semiconductor Device>

(67) The method for manufacturing the semiconductor device in FIGS. 5 to 7 is basically similar to the method for manufacturing the semiconductor device in FIGS. 1 to 3, but the configuration of wiring board 11 and the content of the sealing process (S70) are different from those of the method for manufacturing the semiconductor device in FIGS. 1 to 3. Specifically, sealing member inlet port 13 is formed in wiring board 11 prepared in the preparation process (S10). Sealing member inlet port 13 may be formed at the end portion of wiring board 11. Thereafter, the processes (S20) to (S60) in the method for manufacturing the semiconductor device in FIGS. 1 to 3 are performed. Then, in the sealing process (S70), the resin material or the like as the sealing member is injected into the lower side of wiring board 11 through sealing member inlet port 13. In this way, the semiconductor device in FIGS. 5 to 7 can be obtained.

Advantageous Effect

(68) In the semiconductor device, sealing member inlet port 13 may be formed in wiring board 11. When viewed from the direction perpendicular to main surface 3a of base member 3, sealing member inlet port 13 may be formed at the position overlapping with case 6.

(69) In this case, in addition to the same effect as that of the semiconductor device in FIGS. 1 to 3, sealing member 14 can be easily injected to the lower side of wiring board 11 through sealing member inlet port 13. In addition, distance L2 between conductor layer 24 configuring base member 3 and sealing member inlet port 13 is set larger than thickness T1 of sealing member 14, so that the substances such as the moisture and the gas can be prevented from reaching conductor layer 24 and semiconductor element 4 from the upper surface of sealing member 14 through sealing member inlet port 13.

Third Embodiment

(70) <Configuration of Semiconductor Device>

(71) FIG. 8 is a sectional plan view illustrating a semiconductor device according to a third embodiment. FIG. 9 is a schematic plan view illustrating a positional relationship between the sealing member inlet port and the semiconductor element in the semiconductor device in FIG. 8. FIG. 8 corresponds to FIG. 5, and FIG. 9 corresponds to FIG. 6. The semiconductor device in FIGS. 8 and 9 basically has the same configuration as the semiconductor device in FIGS. 5 to 7, but has the position different from that of sealing member inlet port 13 in wiring board 11. Specifically, sealing member inlet port 13 is formed in the central portion of wiring board 11. In planar view in FIGS. 8 and 9 (when viewed from the direction perpendicular to main surface 3a of base member 3), sealing member inlet port 13 is disposed in the region not overlapping with semiconductor element 4.

(72) Although it is ideal that sealing member inlet port 13 exists at the position farthest from semiconductor element 4, at least a distance L3 between sealing member inlet port 13 and semiconductor element 4 is preferably greater than or equal to thickness T1 (see FIG. 1) of sealing member 14 in planar view of FIG. 9. As described above, sealing member inlet port 13 is not provided on semiconductor element 4, and sealing member inlet port 13 is provided at least at the position separated from semiconductor element 4 by thickness T1 of sealing member 14, so that the gas such as the water vapor or the sulfur gas can be prevented from reaching semiconductor element 4 from the upper surface of sealing member 14 through sealing member inlet port 13.

(73) <Method for Manufacturing Semiconductor Device>

(74) The method for manufacturing the semiconductor device in FIGS. 8 and 9 is basically similar to the method for manufacturing the semiconductor device in FIGS. 1 to 3, but the configuration of wiring board 11 and the content of the sealing process (S70) are different from those of the method for manufacturing the semiconductor device in FIGS. 1 to 3. Specifically, sealing member inlet port 13 is formed in the central portion of wiring board 11 prepared in the preparation process (S10). Thereafter, the processes (S20) to (S60) in the method for manufacturing the semiconductor device in FIGS. 1 to 3 are performed. Then, in the sealing process (S70), the resin material or the like as the sealing member is injected into the lower side of wiring board 11 through sealing member inlet port 13. In this way, the semiconductor device in FIGS. 8 and 9 can be obtained.

Advantageous Effect

(75) In the semiconductor device, sealing member inlet port 13 may be formed in wiring board 11. When viewed from the direction perpendicular to main surface 3a of base member 3, sealing member inlet port 13 may be formed at the position not overlapping with semiconductor element 4.

(76) In this case, in addition to the same effect as that of the semiconductor device in FIGS. 1 to 3, sealing member 14 can be easily injected to the lower side of wiring board 11 through sealing member inlet port 13. In addition, distance L3 between semiconductor element 4 and sealing member inlet port 13 is set larger than thickness T1 of sealing member 14, the substance such as the moisture and the gas can be prevented from reaching semiconductor element 4 from the upper surface of sealing member 14 through sealing member inlet port 13.

Fourth Embodiment

(77) <Configuration of Semiconductor Device>

(78) FIG. 10 is a sectional plan view illustrating a semiconductor device according to a fourth embodiment. FIG. 11 is a schematic sectional view taken along a line XI-XI in FIG. 10. FIG. 10 corresponds to FIG. 1, and FIG. 11 corresponds to FIG. 3. The semiconductor device in FIGS. 10 and 11 basically has the same configuration as the semiconductor device in FIGS. 1 to 3, but the configuration of wiring board 11 and the configurations of support portions 15, 30a, 30b are different from those of the semiconductor device in FIGS. 1 to 3. That is, in the semiconductor device in FIGS. 10 and 11, wiring board 11 includes two boards of a first board 20 and a second board 21. When wiring board 11 includes a plurality of boards, the number of the boards may be at least three.

(79) Support portion **15** and additional support portion **30a**, **30b** protruding in a direction intersecting with the protruding direction of support portion **15** from the side wall surface are formed on the side wall of case **6**. First board **20** is supported by support portion **15** on the left side in FIG. **10** and additional support portion **30a**. Support portion **15** and additional support portion **30a** that are in contact with first board **20** have substantially the same height in the direction perpendicular to main surface **3a** of base member **3**.

(80) The second board **21** is supported by support portion **15** and additional support portion **30b** on the right side in FIG. **10**. Support portion **15** and additional support portion **30b** that are in contact with second board **21** have substantially the same height in the direction perpendicular to main surface **3a** of base member **3**.

(81) In the direction perpendicular to main surface **3a** of base member **3**, additional support portion **30a** is disposed at the position farther from base member **3** than additional support portion **30b**. As can be seen from FIG. **11**, an end portion **20a** of first board **20** is disposed on second board **21** at a distance **L4** from second board **21**. When viewed from the direction perpendicular to main surface **3a** of base member **3**, a part of first board **20** and a part of second board **21** overlap with each other. Distance **L4** is preferably smaller than the width of the overlapping portion between first board **20** and second board **21** in planar view in FIG. **10** (when viewed from the direction perpendicular to main surface **3a** of base member **3**).

(82) A planar shape of each of first board **20** and second board **21** is a quadrangular shape. When the direction along main surface **3a** of base member **3** from first board **20** toward second board **21** is defined as a first direction, and when the direction perpendicular to the first direction and along main surface **3a** is defined as a second direction, in first board **20**, the width in the first direction may be smaller or larger than the width in the second direction. In second board **21**, the width in the first direction may be smaller or larger than the width in the second direction.

(83) <Method for Manufacturing Semiconductor Device>

(84) The method for manufacturing the semiconductor device in FIGS. **10** and **11** is basically similar to the method for manufacturing the semiconductor device in FIGS. **1** to **3**, but is different from the method for manufacturing the semiconductor device in FIGS. **1** to **3** in the configuration of wiring board **11**, the configuration of case **6**, and the content of the process of disposing wiring board **11** (**S50**). Specifically, wiring board **11** prepared in the preparation process (**S10**) is two boards of first board **20** and second board **21**. In addition to support portion **15**, additional support portions **30a**, **30b** are formed on the side wall of case **6**. Thereafter, the processes (**S20**) to (**S40**) in the method for manufacturing the semiconductor device in FIGS. **1** to **3** are performed. Then, in the process of disposing wiring board **11** (**S50**), at first, second board **21** is disposed inside case **6** such that the outer peripheral portion of second board **21** is in contact with support portion **15** and additional support portion **30b**. Subsequently, first board **20** is disposed inside case **6** such that the outer peripheral portion of first board **20** is brought into contact with support portion **15** and additional support portion **30a**. At this time, first board **20** is disposed such that a part of the side of end portion **20a** of the first board **20** overlaps with second board **21**. First board **20** and second board **21** are not in contact with each other and are disposed at distance **L4**. Thereafter, the process (**S60**) and the process (**S70**) in the method for manufacturing the semiconductor device in FIGS. **1** to **3** is performed, so that the semiconductor device in FIGS. **10** and **11** can be obtained.

Advantageous Effect

(85) In the semiconductor device, wiring board **11** may include first board **20** and second board **21**. When viewed from the direction perpendicular to main surface **3a** of base member **3**, a part of first board **20** and a part of second board **21** may overlap with each other. Distance **L4** between first board **20** and second board **21** in the region where first board **20** and second board **21** overlap with each other is preferably smaller than the width of the overlapping portion between first board **20** and second board **21** in planar view in FIG. **10** (when viewed from the direction perpendicular to main surface **3a** of base member **3**).

(86) In this case, the effects similar to those of the semiconductor device in FIGS. 1 to 3 can be obtained, and the possibility that the substance such as the moisture and the gas reaches semiconductor element 4 from the upper surface of sealing member 14 through the overlapping portion between first board 20 and second board 21 can be decreased.

(87) <Configuration and Advantageous Effect of Modification>

(88) FIG. 12 is a schematic sectional view illustrating a first modification of the semiconductor device of the fourth embodiment. FIG. 12 corresponds to FIG. 11. The semiconductor device in FIG. 12 basically has the same configuration as the semiconductor device in FIGS. 10 and 11, but is different from the semiconductor device in FIGS. 10 and 11 in that first board 20 and second board 21 are in contact with each other in the region where first board 20 and second board 21 overlap with each other.

(89) In this case, the effects similar to those of the semiconductor device in FIGS. 10 and 11 can be obtained. In the semiconductor device in FIG. 12, first board 20 and second board 21 are in contact with each other, so that the possibility that the substance such as the moisture and the gas reaches semiconductor element 4 from the upper surface of sealing member 14 through the overlapping region can be decreased.

(90) FIG. 13 is a schematic plan view illustrating a second modification of the semiconductor device of the fourth embodiment. The semiconductor device in FIG. 13 basically has the same configuration as that of the semiconductor device in FIGS. 10 and 11, but is different from the semiconductor device in FIGS. 10 and 11 in that first board 20 and second board 21 are disposed at intervals without overlapping with each other in planar view. When viewed from the direction perpendicular to main surface 3a of base member 3, first board 20 and second board 21 are disposed at an interval. End portion 20a of first board 20 and an end portion 21a of second board 21 face each other at an interval. When viewed from the above direction, semiconductor element 4 is disposed at a position not overlapping with a gap 31 between first board 20 and second board 21. A distance L5 between gap 31 and semiconductor element 4 as viewed from the above direction is preferably larger than thickness T1 (see FIG. 1) of sealing member 14 in the above direction.

(91) Summarizing the characteristic configuration of the semiconductor device described above, wiring board 11 may include first board 20 and second board 21. When viewed from the direction perpendicular to main surface 3a of base member 3, first board 20 and second board 21 may be disposed at an interval. When viewed from the above direction, semiconductor element 4 may be disposed at the position not overlapping with gap 31 between first board 20 and second board 21. Distance L5 between gap 31 and semiconductor element 4 as viewed from the above direction may be larger than thickness T1 of sealing member 14 in the above direction.

(92) In this case, distance L5 from gap 31 to semiconductor element 4 is sufficiently secured, so that the possibility that the substance such as the moisture or the gas reaches semiconductor element 4 from the upper surface of sealing member 14 through gap 31 can be decreased.

Fifth Embodiment

(93) <Configuration of Semiconductor Device>

(94) FIG. 14 is a sectional plan view illustrating a semiconductor device according to a fifth embodiment. FIG. 15 is a schematic sectional view taken along a line XV-XV in FIG. 14. FIG. 16 is a schematic sectional view taken along a line XVI-XVI in FIG. 14. FIGS. 14 to 16 correspond to FIGS. 1 to 3, respectively. The semiconductor device in FIGS. 14 to 16 basically has the same configuration as the semiconductor device in FIGS. 1 to 3, but is different from the semiconductor device in FIGS. 1 to 3 in the configuration of base member 3 and heat dissipation layer 1 as the separate member from base member 3, and that case 6 and base member 3 are connected to each other with heat dissipation layer 1 interposed therebetween. That is, in the semiconductor device in FIGS. 14 to 16, base member 3 includes the insulating layer made of an inorganic ceramic material and the conductor layers connected to the upper surface and the lower surface of the insulating layer. For example, alumina (Al.sub.2O.sub.3), aluminum nitride (AlN), silicon nitride

(Si.sub.3N.sub.4), silicon dioxide (SiO.sub.2), or boron nitride (BN) can be used as the inorganic ceramic material. The size of base member **3** is smaller than the size of wiring board **11**.

(95) Heat dissipation layer **1** is connected to the conductor layer configuring the lower surface of base member **3** with bonding material **2** such as solder interposed therebetween. Case **6** is connected to the outer peripheral portion of heat dissipation layer **1** with the adhesive (not illustrated) interposed therebetween.

(96) <Method for Manufacturing Semiconductor Device>

(97) The method for manufacturing the semiconductor device in FIGS. **14** to **16** is basically similar to the method for manufacturing the semiconductor device in FIGS. **1** to **3**, but is different from the method for manufacturing the semiconductor device in FIGS. **1** to **3** in the configurations of base member **3** and heat dissipation layer **1** and the content of the process (S30) of bonding case **6** to base member **3**. Specifically, a multilayer body in which the insulating layer made of an inorganic ceramic material as described above and two conductor layers are laminated is prepared as base member **3** prepared in the preparation process (S10). Heat dissipation layer **1** is connected to base member **3** with bonding material **2** interposed therebetween.

(98) Thereafter, the process (S20) in the method for manufacturing the semiconductor device in FIGS. **1** to **3** is performed. Then, in the process (S30), case **6** is bonded to the outer peripheral portion of heat dissipation layer **1** using the adhesive. Thereafter, the processes (S40) to (S70) in the method for manufacturing the semiconductor device in FIGS. **1** to **3** are performed. In this way, the semiconductor device in FIGS. **14** to **16** can be obtained.

Advantageous Effect

(99) The semiconductor device in FIGS. **14** to **16** may include heat dissipation layer **1** connected to base member **3**. In this case, the effect similar to that of the semiconductor device in FIGS. **1** to **3** can be obtained, and heat dissipation layer **1** and base member **3** are prepared as the separate members, so that a degree of freedom in selecting the configuration of the semiconductor device can be increased.

Sixth Embodiment

(100) In the present embodiment, the semiconductor device of any one of the first to fifth embodiments described above is applied to a power conversion device. Although the present disclosure is not limited to a specific power conversion device, the case that the present disclosure is applied to a three-phase inverter will be described below as the sixth embodiment.

(101) FIG. **17** is a block diagram illustrating a configuration of a power conversion system to which the power conversion device according to the present embodiment is applied.

(102) The power conversion system in FIG. **17** includes a power supply **100**, a power conversion device **200**, and a load **300**. Power supply **100** is a DC power supply, and supplies DC power to power conversion device **200**. Power supply **100** can be configured by various components. For example, power supply **100** can be configured by a DC system, a solar cell, and a storage battery, or may be configured by a rectifier circuit connected to an AC system, an AC-DC converter, and the like. Power supply **100** may be constructed with a DC-DC converter that converts the DC power output from the DC system into predetermined power.

(103) Power conversion device **200** is a three-phase inverter connected between power supply **100** and load **300**, converts the DC power supplied from power supply **100** into AC power, and supplies the AC power to load **300**. As illustrated in FIG. **17**, power conversion device **200** includes a main conversion circuit **201** that converts the DC power into the AC power to output the AC power and a control circuit **203** that outputs a control signal controlling main conversion circuit **201** to main conversion circuit **201**.

(104) Load **300** is a three-phase motor driven by the AC power supplied from power conversion device **200**. Load **300** is not limited to a specific application, but is a motor mounted on various electric appliances. For example, load **300** is used as a hybrid car, an electric car, a rail vehicle, an elevator, or a motor for an air conditioner.

(105) Power conversion device **200** will be described in detail below. Main conversion circuit **201** includes a switching element and a freewheel diode (not illustrated), converts the DC power supplied from power supply **100** into the AC power by switching of the switching element, and supplies the AC power to load **300**. Although there are various specific circuit configurations of main conversion circuit **201**, main conversion circuit **201** of the present embodiment is a two-level three-phase full bridge circuit, and can be configured by six switching elements and six freewheel diodes connected in anti-parallel to the respective switching elements. At least one of each switching element and each freewheel diode of main conversion circuit **201** is a switching element or a freewheel diode included in semiconductor device **202** corresponding to the semiconductor device of at least one of the first to fifth embodiments. Six switching elements are connected in series in every two switching elements to constitute upper and lower arms, and each upper and lower arm constitutes each phase (U-phase, V-phase, W-phase) of the full bridge circuit. An output terminal of each of the upper and lower arms, namely, three output terminals of main conversion circuit **201** are connected to load **300**.

(106) Main conversion circuit **201** includes a drive circuit (not illustrated) that drives each switching element, but the drive circuit may be built in semiconductor device **202**, or include the drive circuit separately from semiconductor device **202**. The drive circuit generates a drive signal driving the switching element of main conversion circuit **201**, and supplies the drive signal to the control electrode of the switching element of main conversion circuit **201**. Specifically, the drive signal turning on the switching element and the drive signal turning off the switching element are output to the control electrode of each switching element according to the control signal from control circuit **203** (described later). The drive signal is a voltage signal (ON signal) greater than or equal to a threshold voltage of the switching element when the switching element is maintained in an ON state, and the drive signal is a voltage signal (OFF signal) equal to or smaller than the threshold voltage of the switching element when the switching element is maintained in an OFF state.

(107) Control circuit **203** controls the switching elements of main conversion circuit **201** so that desired power is supplied to load **300**. Specifically, time (ON time) during which each switching element of main conversion circuit **201** is to be turned on is calculated based on the power to be supplied to load **300**. For example, main conversion circuit **201** can be controlled by PWM control that modulates the ON time of the switching element according to the voltage to be output. A control command (control signal) is output to the drive circuit included in main conversion circuit **201** such that the ON signal is output to the switching element to be turned on at each time point, and such that the OFF signal is output to the switching element to be turned off at each time point. The drive circuit outputs the ON signal or the OFF signal as the drive signal to the control electrode of each switching element according to the control signal.

(108) In the power conversion device of the present embodiment, the semiconductor device of any one of the first to fifth embodiments is applied as semiconductor device **202** configuring main conversion circuit **201**, so that the reliability can be improved.

(109) Although the example in which the present disclosure is applied to the two-level three-phase inverter is described in the present embodiment, the present disclosure is not limited to the present embodiment, and can be applied to various power conversion devices. In the present embodiment, the two-level power conversion device is used. However, a three-level or multi-level power conversion device may be used, or the present disclosure may be applied to a single-phase inverter when the power is supplied to a single-phase load. In addition, the present disclosure can also be applied to a DC/DC converter, an AC-DC converter, or the like when the power is supplied to a DC load or the like.

(110) In addition, the power conversion device to which the present disclosure is applied is not limited to the case where the load described above is the electric motor, and can also be used as, for example, a power supply device for an electric discharge machine, a laser beam machine, an

induction heating cooker, or a non-contact power feeding system, and can also be used as a power conditioner for a solar power generation system, a power storage system, or the like.
(111) It should be considered that the disclosed embodiments are an example in all respects and not restrictive. As long as there is no contradiction, at least two of the disclosed embodiments may be combined. The basic scope of the present disclosure is defined by not the above description but the claims, and it is intended that all modifications within the meaning and scope of the claims are included in the present invention.

REFERENCE SIGNS LIST

(112) **1**: heat dissipation layer, **2**: bonding material, **3**: base member, **3a**: main surface, **4**: semiconductor element, **5**: conductive wire, **6**: case, **6a**: case surface, **7**: external connection terminal, **8**: internal connection terminal, **9**: terminal member, **9a**, **20a**, **21a**: end portion, **10**: electronic component, **11**: wiring board, **11a**, **11c**: surface, **11aa**: contact region, **11b**: outer peripheral edge, **12**: through-hole, **13**: sealing member inlet port, **14**: sealing member, **15**, **30a**, **30b**: support portion, **15a**: notch, **15b**: recessed portion, **15c**: support surface, **15ca**: support surface end portion, **20**: first board, **21**: second board, **24**: conductor layer, **25**: insulating layer, **31**: gap, **61**: side wall, **61a**: inner circumferential surface, **100**: power supply, **200**: electric power conversion device, **201**: main conversion circuit, **202**: semiconductor device, **203**: control circuit, **300**: load

Claims

1. A semiconductor device comprising: a base member including a main surface; a semiconductor element mounted on the main surface of the base member; a case including a side wall surrounding the base member, a support portion being formed on an inner peripheral surface of the side wall on a base member side; a wiring board disposed at a position overlapping with the semiconductor element when viewed from a direction perpendicular to the main surface of the base member, an electronic component being mounted on the wiring board, at least a part of the support portion being in contact with a contact region located inside an outer peripheral edge of the wiring board on a surface of the wiring board located on a semiconductor element side; a sealing member disposed inside the case to seal the semiconductor element and the wiring board; and a terminal member connected to a region on an outer peripheral edge side with respect to the contact region in the wiring board and including an end portion protruding from a surface of the sealing member, wherein at least a part of the support portion includes a recessed portion facing a region to which the terminal member is connected in the wiring board.
2. The semiconductor device according to claim 1, wherein an internal connection terminal electrically connected to the wiring board is disposed inside the contact region in contact with the support portion on the surface of the wiring board located on the semiconductor element side.
3. The semiconductor device according to claim 1, wherein a water vapor transmission rate of a material constituting the wiring board and the case is smaller than a water vapor transmission rate of the sealing member.
4. The semiconductor device according to claim 1, wherein the base member includes a conductor layer to which the semiconductor element is connected, and a size of a surface of the wiring board on which the electronic component is mounted is larger than a size of the conductor layer.
5. The semiconductor device according to claim 1, wherein a sealing member inlet port is formed in the wiring board, and the sealing member inlet port is formed at a position overlapping with the case when viewed from the direction perpendicular to the main surface of the base member.
6. The semiconductor device according to claim 1, wherein a sealing member inlet port is formed in the wiring board, and the sealing member inlet port is formed at a position not overlapping with the semiconductor element when viewed from the direction perpendicular to the main surface of the base member.
7. The semiconductor device according to claim 1, wherein the wiring board includes a first board

and a second board, and a part of the first board and a part of the second board overlap with each other when viewed from the direction perpendicular to the main surface of the base member.

8. The semiconductor device according to claim 1, wherein the wiring board includes a first board and a second board, the first board and the second board are disposed at an interval when viewed from the direction perpendicular to the main surface of the base member, the semiconductor element is disposed at a position not overlapping with a gap between the first board and the second board when viewed from the direction, and a distance between the gap and the semiconductor element as viewed from the direction is larger than a thickness of the sealing member in the direction.

9. The semiconductor device according to claim 1, wherein the base member includes a heat dissipation layer, and a material constituting the heat dissipation layer is metal.

10. The semiconductor device according to claim 1, further comprising a heat dissipation layer connected to the base member.

11. A power conversion device comprising: a main conversion circuit including the semiconductor device according to claim 1, and to convert and output input power; and a control circuit to output a control signal controlling the main conversion circuit to the main conversion circuit.
